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Program Resources

LESSON 3-5

Zeros of Polynomial Functions

Lesson Overview

FOCUS

Objective

Students will be able to:

✔ Identify the zeros of a function by factoring or using synthetic division.

✔ Use the zeros of a polynomial function to sketch its graph.

Essential Understanding

The zeros of a polynomial function can be determined using factoring or synthetic division. The zeros of a function can be used to sketch its graph.

COHERENCE

Previously in this course, students:

- Factored polynomials using polynomial identities.
- Divided polynomials using synthetic division.

In this lesson, students:

- Use factoring or synthetic division to identify the zeros of a polynomial function.
- Use the zeros of a polynomial to sketch a graph of a function defined by the polynomial.

Later in this topic, students will:

- Find the possible roots of a polynomial equation using graphing, substitution, or synthetic division.
- Use the Rational Root Theorem and the Conjugates Theorem to determine which of the possible roots are zeros.

RIGOR

This lesson emphasizes a blend of *procedural skill and fluency* and *application*.

- Students use factoring or synthetic division to find the zeros of a polynomial function.
- Students interpret the key features of the graph of a polynomial function in the context of problems about profit models.

Glossary is available at SavvasRealize.com.

Vocabulary Builder

REVIEW VOCABULARY

English | *Spanish*

- **extraneous solution** | *solucion extraña*
- **zero of a function** | *cero de una función*
- **Zero Product Property** | *propiedad del product de cero*

NEW VOCABULARY

- **multiplicity of a zero** | *multiplicidad de cero*

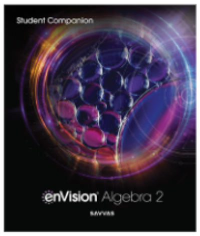
VOCABULARY ACTIVITY

Review the term *zero of a function* by asking students to describe what it means in the context of a quadratic equation and its graph. Explain that a polynomial function can have a factor that occurs twice. Introduce the term *multiplicity of a zero* and explain what it means. Have students match each factor to the zero.

$(x-2)$	4
$(x+4)$	-4
$(x+2)$	-2
$(x-4)$	2

Student Companion

Students can do their work for the lesson in their *Student Companion* or on Savvas Realize.



Mathematics Overview

Students use factoring and synthetic division to find the zeros of a polynomial function and determine how the multiplicity of a zero affects the graph of the function.

Students use the zeros of a function, including the multiplicity of each zero and whether the zero is real or complex, to sketch the graph of the polynomial function.

Applying Math Practices

Look For and Make Use of Structure

Students make connections between mathematical concepts when they recognize that statements about zeros, factors, and x-intercepts are all ways of describing the same relationship.

Look For and Express Regularity in Repeated Reasoning

Students generalize the relationship between the multiplicity of zeros and the appearance of the graph of the function it defines.

MODEL & DISCUSS

INSTRUCTIONAL FOCUS Students consider the height of a rocket relative to time, building their understanding of how the zeros of a function can be interpreted within the context of real-world problems.

STUDENT COMPANION Students can complete the *Model & Discuss* activity in their *Student Companion*.

Before WHOLE CLASS

Implement Tasks that Promote Reasoning and Problem Solving ETP

Q: How would you describe the function that models the height of the rocket?
[It is a quadratic function written in standard form. The variable t represents the time in seconds after the rocket was launched. The value of the function represents the height of the rocket t seconds after it was launched.]

During SMALL GROUP

Support Productive Struggle in Learning Mathematics ETP

Q: Why do you need additional information to construct an accurate model for the height of the rocket?
[You need to know three points to write the equation of a parabola. You are only given two points in the problem, the launch point and the landing point.]

For Early Finishers

Q: Another rocket was launched that fell to the ground 10 seconds after it was launched. How does the maximum height of this rocket compare to the first rocket?
[The maximum height of this rocket is four times as high as that of the first rocket.]

After WHOLE CLASS

Facilitate Meaningful Mathematical Discourse ETP

Discuss the accuracy of Charlie's model.

Q: What information in the problem statement helps you determine the value of c in the model?
[The rocket is launched from the ground. So when $t = 0$, $h(0) = 0$.]

Q: How can you determine the maximum height of the rocket in Charlie's model?
[The maximum height occurs at $t = 2.5$, halfway between 0 and 5. Evaluate the function when $t = 2.5$ to find the maximum height.]

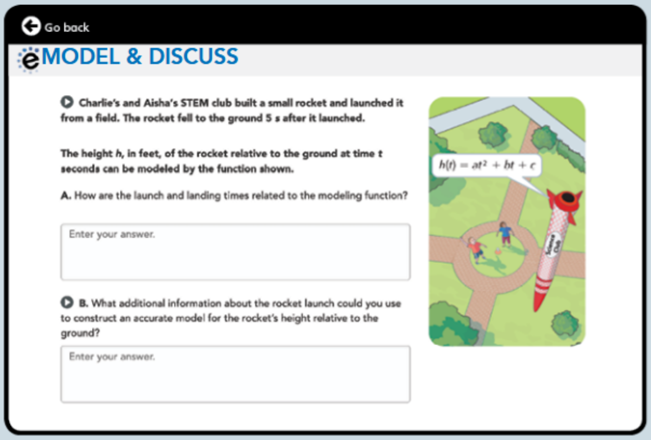
Use with **MODEL & DISCUSS**

HABITS OF MIND

Reason In Charlie's function, what is the value of c? Why is this the correct value?

[0; Because the rocket is launched from the ground, or a height of 0 ft, the value of c is zero.]

Critique & Explain is available at SavvasRealize.com.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MODEL & DISCUSS

Charlie's and Aisha's STEM club built a small rocket and launched it from a field. The rocket fell to the ground 5 s after it launched.

The height h, in feet, of the rocket relative to the ground at time t seconds can be modeled by the function shown.

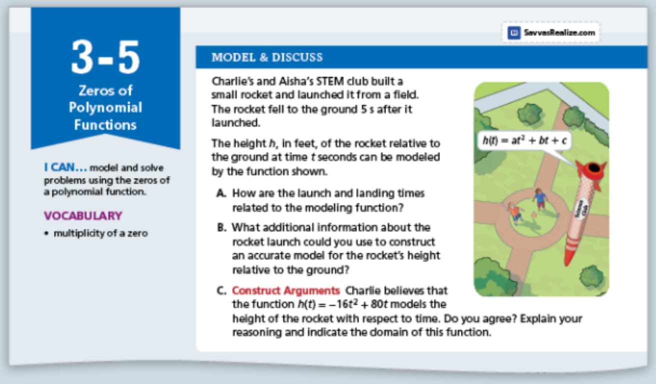
A. How are the launch and landing times related to the modeling function?

Enter your answer.

B. What additional information about the rocket launch could you use to construct an accurate model for the rocket's height relative to the ground?

Enter your answer.

STUDENT EDITION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3-5 Zeros of Polynomial Functions

I CAN… model and solve problems using the zeros of a polynomial function.

VOCABULARY
- multiplicity of a zero

MODEL & DISCUSS

Charlie's and Aisha's STEM club built a small rocket and launched it from a field. The rocket fell to the ground 5 s after it launched.

The height h, in feet, of the rocket relative to the ground at time t seconds can be modeled by the function shown.

A. How are the launch and landing times related to the modeling function?

B. What additional information about the rocket launch could you use to construct an accurate model for the rocket's height relative to the ground?

C. Construct Arguments Charlie believes that the function $h(t) = -16t^2 + 80t$ models the height of the rocket with respect to time. Do you agree? Explain your reasoning and indicate the domain of this function.

SAMPLE STUDENT WORK

A. The height of the rocket is 0 at both the launch and landing. So 0 seconds and 5 seconds are the values of t that solve the equation $0 = at^2 + bt + c$

B. You need information about the height of the rocket at another time, knowledge about the rocket's initial speed and about gravitational acceleration.

C. Yes. Charlie's model is 0 at both 0 seconds and 5 seconds. The domain is all x-values from 0 to 5.

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Activity

INTRODUCE THE ESSENTIAL QUESTION

Establish Mathematics Goals to Focus Learning ETP

Remind students that they have previously found the zeros of linear and quadratic functions. Explain that they can use factoring, synthetic division, and the Quadratic Formula to find the zeros of polynomial functions.

EXAMPLE 1 Use Zeros to Graph a Polynomial Function

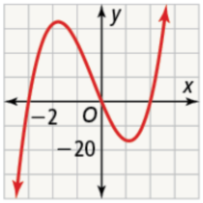
Pose Purposeful Questions ETP

Q: How is the Zero Product Property helpful when graphing a polynomial function?

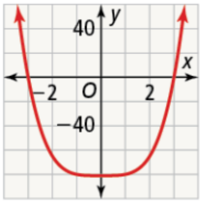
[It helps you identify ordered pairs that are on the graph and determine endpoints of intervals.]

Try It! Answers

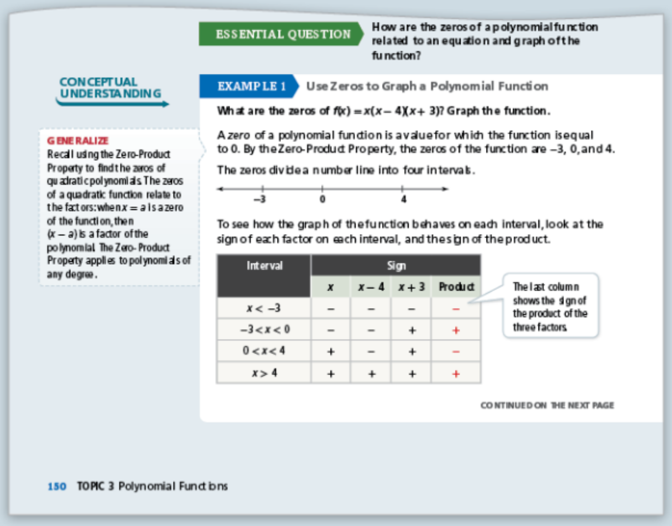
1. a. Zeros are −3, 0, and 2.



b. Zeros are −3 and 3.



Examples are available at SavvasRealize.com.



ESSENTIAL QUESTION How are the zeros of a polynomial function related to an equation and graph of the function?

CONCEPTUAL UNDERSTANDING

EXAMPLE 1 Use Zeros to Graph a Polynomial Function

What are the zeros of $f(x) = x(x - 4)(x + 3)$? Graph the function.

A *zero* of a polynomial function is a value for which the function is equal to 0. By the Zero-Product Property, the zeros of the function are −3, 0, and 4.

The zeros divide a number line into four intervals.

−3 0 4

To see how the graph of the function behaves on each interval, look at the sign of each factor on each interval, and the sign of the product.

Interval	Sign			
	x	$x - 4$	$x + 3$	Product
$x < -3$	−	−	−	−
$-3 < x < 0$	−	−	+	+
$0 < x < 4$	+	−	+	−
$x > 4$	+	+	+	+

The last column shows the sign of the product of the three factors.

GENERALIZE Recall using the Zero-Product Property to find the zeros of quadratic polynomials. The zeros of a quadratic function relate to the factors: when $x = a$ is a zero of the function, then $(x - a)$ is a factor of the polynomial. The Zero-Product Property applies to polynomials of any degree.

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Elicit and Use Evidence of Student Thinking ETP

Q: What step is important before sketching the graph of each function?

[Factor each function to find the zeros of the function.]

Additional Examples

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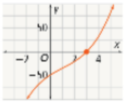
ADDITIONAL EXAMPLE 1 ESSENTIAL QUESTION

Sketch the graph of the function by finding the zeros.

$f(x) = x^3 - 3x^2 + 16x - 48$

$f(x) = x^3 - 3x^2 + 16x - 48$
$= (x - 3)(x^2 + 16)$

Zero at $x = 3$



SOLUTION

Additional Examples are available at SavvasRealize.com.

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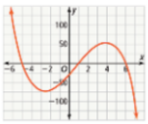
ADDITIONAL EXAMPLE 4 ESSENTIAL QUESTION

A company manufactures and sells Adirondack chairs. The company's profit P, in thousands of dollars earned, is a function of the number of chairs sold, x, measured in thousands. The profit is modeled by the function:

$P(x) = -x^3 + 2x^2 + 29x - 30$

What do the zeros of the function tell you about the number of chairs that the company should produce?



Based on the graph of the function, the zeroes are at −5, 1, and 6.

When $P(x)$ is positive, the company earns a profit. The function is positive on $x < -5$, and $1 < x < 6$.

Since the company cannot produce a negative amount of chairs, disregard values for x less than −5.

So the company should make between 1,000 and 6,000 chairs to make a profit.

SOLUTION

Example 1 Students sketch the graph of a cubic function with only one real zero.

Q: How does the number of real zeros affect the shape of the graph?

[The graph of the function only crosses the x-axis one time.]

Example 4 Students interpret the positive zeros of a polynomial function.

Q: What do the values of the positive x-intercepts represent in this context?

[The values represent the break-even points where the company would neither lose money nor make a profit.]

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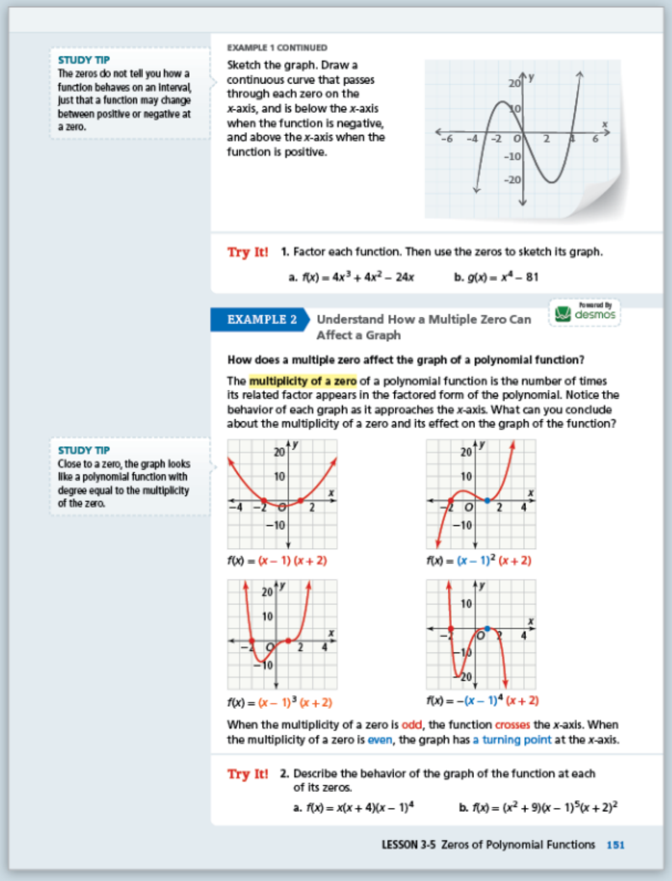
Activity

EXAMPLE 2 Understand How a Multiple Zero Can Affect a Graph

Example 2 is Powered by Desmos.
Examples are available at SavvasRealize.com.



STUDY TIP
The zeros do not tell you how a function behaves on an interval, just that a function may change between positive or negative at a zero.

EXAMPLE 1 CONTINUED

Sketch the graph. Draw a continuous curve that passes through each zero on the x-axis, and is below the x-axis when the function is negative, and above the x-axis when the function is positive.

Try It! 1. Factor each function. Then use the zeros to sketch its graph.

a. $f(x) = 4x^3 + 4x^2 - 24x$ b. $g(x) = x^4 - 81$

EXAMPLE 2 Understand How a Multiple Zero Can Affect a Graph

Powered By desmos

How does a multiple zero affect the graph of a polynomial function?

The **multiplicity of a zero** of a polynomial function is the number of times its related factor appears in the factored form of the polynomial. Notice the behavior of each graph as it approaches the x-axis. What can you conclude about the multiplicity of a zero and its effect on the graph of the function?

STUDY TIP
Close to a zero, the graph looks like a polynomial function with degree equal to the multiplicity of the zero.

$f(x) = (x - 1)(x + 2)$

$f(x) = (x - 1)^2(x + 2)$

$f(x) = (x - 1)^3(x + 2)$

$f(x) = -(x - 1)^4(x + 2)$

When the multiplicity of a zero is odd, the function crosses the x-axis. When the multiplicity of a zero is even, the graph has a turning point at the x-axis.

Try It! 2. Describe the behavior of the graph of the function at each of its zeros.

a. $f(x) = x(x + 4)(x - 1)^4$ b. $f(x) = (x^2 + 9)(x - 1)^5(x + 2)^2$

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Use and Connect Mathematical Representations ETP

Q: What do you notice about each graph in relation to the polynomial function it represents?
[The number of unique factors of the polynomial function is the same as the number of times that its graph crosses or is tangent to the x-axis.]

Q: How does the term *multiplicity of a zero* relate to multiplication?
[The multiplicity of a zero is the number of times that the factor is multiplied in the polynomial function.]

Try It! Answers

2. a. The graph crosses the x-axis at the zeros -4 and 0 and has a turning point at the zero 1

b. The graph crosses the x-axis at the zero 1, and has a turning point at the zero -2.

Elicit and Use Evidence of Student Thinking ETP

Q: What do you know about the signs of the y-values of the points in the intervals on each side of a zero whose multiplicity is even?
[The signs are the same.]

HABITS OF MIND

Use with EXAMPLES 1 & 2

Reason Do the values of a function always change from positive to negative or negative to positive on either side of a zero? Explain.

[No; the zero could be a turning point, meaning the function is positive or negative on both sides of the zero.]

ELL English Language Learners (*Use with* EXAMPLE 2)

SPEAKING BEGINNING Have students discuss the meanings of *odd* and *even*. Then discuss ways to remember how the number of times a zero is repeated affects the graph of a polynomial function.

Q: What does it mean for a number to be *even*? *Odd*?
[divisible by 2; not divisible by 2]

Q: Name one way to remember that when a zero is repeated an *odd* number of times, the graph crosses over the x-axis.
[*Odd* and *over* both start with *o*.]

Q: Name one way to remember that when a zero is repeated an *even* number of times, the graph does not cross the x-axis.
[*Even* ends with an *n* for *not cross*.]

READING DEVELOPING Ask students to read the example and answer the questions to demonstrate their understanding.

Q: In each function, what is the multiplicity of -2? [1]

Q: What happens to each graph at $x = -2$?
[It crosses the x-axis.]

Q: In which functions is the multiplicity of 1 odd? [the two functions on the left]

Q: What happens to the graphs at $x = 1$ when the multiplicity of 1 is odd?
[It crosses the x-axis.]

Q: In which functions is the multiplicity of 1 even? [the two functions on the right]

Q: What happens to the graphs at $x = 1$ when the multiplicity of 1 is even?
[It is tangent to the x-axis.]

LISTENING EXPANDING *Multiplicity* is not commonly used in everyday conversation. But it is related to *multiple* and *multiply*, which are more commonly used.

Q: In your journals, write definitions for *multiple, multiply,* and *multiplicity* in your own words and use each term in a sentence.
[Check students' work.]

Q: Write a paragraph using $(x + 4)$, $(x - 1)$, *multiple*, and *multiply* to explain the meaning of *multiplicity*.
[In the function $f(x) = (x + 4)^2(x - 1)^3$, the factors $(x + 4)$ and $(x - 1)$ occur *multiple* times. The factor $(x + 4)$ is *multiplied* by itself, so the *multiplicity* of -4 is two. The factor $(x - 1)$ occurs three times, so the *multiplicity* of 1 is three.]

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Activity

EXAMPLE 3 Find Real and Complex Zeros

Pose Purposeful Questions ETP

Q: What do you notice about the graph that helps in determining the zeros of the function?
[The graph only crosses the x-axis once and does not have a turning point on the x-axis, so there is only one real zero, and the multiplicity of that zero is odd.]

Q: Why is it helpful to use synthetic division to find a zero of a function?
[If you use synthetic division, and there is no remainder, then the number you divided by is a zero of the function.]

Try It! Answers

3. a. $x = 3$ and $x = \frac{(1 \pm i\sqrt{5})}{2}$
b. $x = -2$, $x = 2$, $x = i$, $x = -i$

Elicit and Use Evidence of Student Thinking ETP

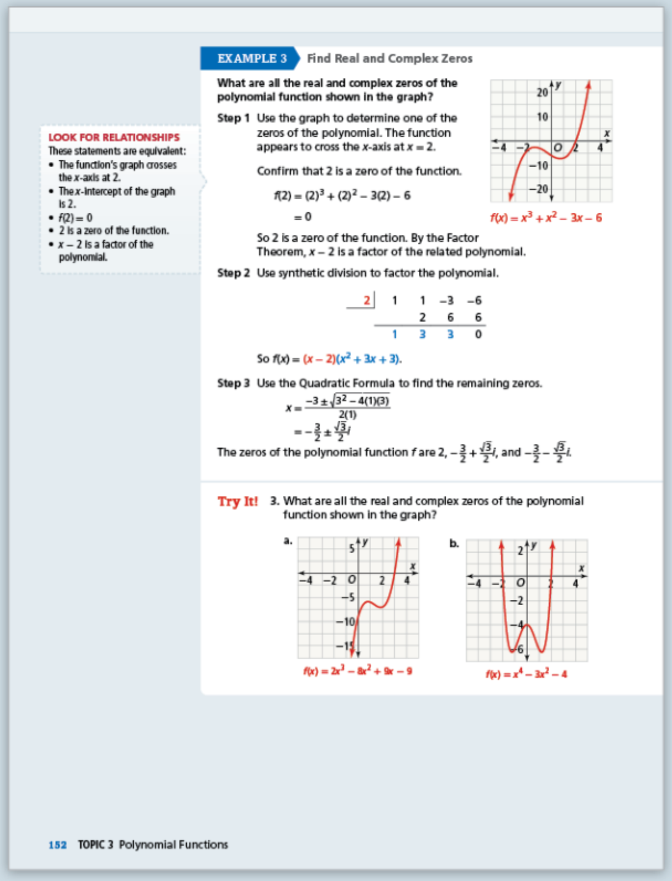
Q: What do you notice about the graph of the function in part (a)?
[The function crosses the x-axis once. It appears to have a zero at $x = 3$.]

Q: Are there any types of functions where the function values change signs without intersecting the x-axis?
[Yes, a function that is not continuous, such as a piecewise step function, can change signs without having a zero on the x-axis.]

Common Error

Try It! 3a Students may state that $x = 3$ is the only zero of the function. Remind students that the graph only shows real zeros, so they also need to find the complex zeros of the function using the Quadratic Formula.

Examples are available at SavvasRealize.com.



EXAMPLE 3 Find Real and Complex Zeros

LOOK FOR RELATIONSHIPS
These statements are equivalent:
- The function's graph crosses the x-axis at 2.
- The x-intercept of the graph is 2.
- $f(2) = 0$
- 2 is a zero of the function.
- $x - 2$ is a factor of the polynomial.

What are all the real and complex zeros of the polynomial function shown in the graph?

$f(x) = x^3 + x^2 - 3x - 6$

Step 1 Use the graph to determine one of the zeros of the polynomial. The function appears to cross the x-axis at $x = 2$.

Confirm that 2 is a zero of the function.

$$f(2) = (2)^3 + (2)^2 - 3(2) - 6$$
$$= 0$$

So 2 is a zero of the function. By the Factor Theorem, $x - 2$ is a factor of the related polynomial.

Step 2 Use synthetic division to factor the polynomial.

$$\begin{array}{r|rrrr} 2 & 1 & 1 & -3 & -6 \\ & & 2 & 6 & 6 \\ \hline & 1 & 3 & 3 & 0 \end{array}$$

So $f(x) = (x - 2)(x^2 + 3x + 3)$.

Step 3 Use the Quadratic Formula to find the remaining zeros.

$$x = \frac{-3 \pm \sqrt{3^2 - 4(1)(3)}}{2(1)}$$
$$= -\frac{3}{2} \pm \frac{\sqrt{3}}{2}i$$

The zeros of the polynomial function f are 2, $-\frac{3}{2} + \frac{\sqrt{3}}{2}i$, and $-\frac{3}{2} - \frac{\sqrt{3}}{2}i$.

Try It! 3. What are all the real and complex zeros of the polynomial function shown in the graph?

a. $f(x) = 2x^3 - 8x^2 + 9x - 9$

b. $f(x) = x^4 - 3x^2 - 4$

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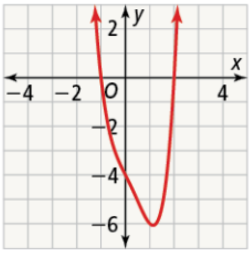
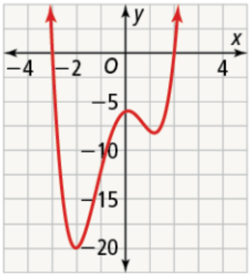
Extend Student Thinking

USE WITH EXAMPLE 3 Have students find all the real and complex zeros of fourth-degree polynomials.

1. Find all the real and complex zeros of $f(x) = x^4 - x^3 - 2x - 4$.
[-1, 2, $\pm i\sqrt{2}$]



2. Find all the real and complex zeros of $f(x) = x^4 + x^3 - 5x^2 + x - 6$.
[-3, 2, $\pm i$]



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Activity

EXAMPLE 4 Find the Polynomial Model

Examples are available at SavvasRealize.com.

Implement Tasks that Promote Reasoning and Problem Solving ETP

Q: Why is it important to think about the domain of the function before starting?
[Because this is a real-life example, there may be values that mathematically satisfy the equation, but do not make sense in the context of the problem.]

Q: Why would only discrete values greater than 0 be in the domain of this function?
[The context of this function is selling lamps. You cannot sell partial lamps or negative lamps, so only values greater than 0 that represent a whole number of lamps make sense in this problem.]

Q: What do the zeros represent in the context of the problem?
[The zeros $x = 3$ and $x = 10$ are points where the company's profit is 0. The zero $x = -2$ has no meaning in this context because the company cannot sell a negative number of lamps.]

Try It! Answer

4. between 1,000 and 11,000

Elicit and Use Evidence of Student Thinking ETP

Q: How do you think the new maximum profit will compare to the maximum profit in the example?
[Because the cost of the materials is decreasing, the maximum profit will be greater.]

Use with EXAMPLES 3 & 4

HABITS OF MIND

Make Sense and Persevere On a graph, how do complex roots differ from real roots?

[Real roots lie on the x-axis, while complex roots do not.]

APPLICATION

EXAMPLE 4 Find the Polynomial Model

Acme Innovations makes and sells lamps. Their profit P, in hundreds of dollars earned, is a function of the number of lamps sold x, in thousands.

From historical data, they know that their company's profit is modeled by the function shown.

The function has zeros at -2, 3, and 10. When Acme Innovations sells 2,000 lamps, they lose \$3,200. Find P in standard form and interpret the key features for this situation.

Formulate: A profit for the company corresponds to the portions of the graph that lie in the first quadrant. Use the zeros of the function to determine the factors of P.

By the Factor Theorem you can determine the factors of the related polynomial.

Zero of P	Factor of $P(x)$
-2	$x + 2$
3	$x - 3$
10	$x - 10$

By the end behavior, the polynomial's leading coefficient is negative, and the function's degree is odd. Since there are three distinct zeros, the least degree the polynomial could have is 3.

Compute: Multiply these factors and -1 to find the standard form of the polynomial.

$$-(x+2)(x-3)(x-10) = (-x-2)(x^2-13x+30)$$
$$= -x^3 + 13x^2 - 30x - 2x^2 + 26x - 60$$
$$= -x^3 + 11x^2 - 4x - 60$$

REASON
Before multiplying to find the standard form of the function, what could you do to determine the multiplicity of the factors or a stretch factor?

Find $P(2)$ to see if polynomial goes through $(2, -32)$.

$$P(2) = -(2)^3 + 11(2)^2 - 4(2) - 60$$
$$= -8 + 44 - 8 - 60$$
$$= -32$$

So $P(x) = -x^3 + 11x^2 - 4x - 60$.

Interpret: From the y-intercept, Acme Innovations spends \$6,000 to set up production before they make the first lamp. When they make and sell 3,000 or 10,000 lamps, they break even. They make a profit when they make and sell between those values. Using technology, you can find they earn the most when they make and sell 7,147 lamps for a profit of \$10,822.

Try It! 4. Due to a decrease in the cost of materials, the profit function for Acme Innovations has changed to $Q(x) = -x^3 + 10x^2 + 13x - 22$. How many lamps should they make in order to make a profit?

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Activity

EXAMPLE 5 Solve Polynomial Equations

Build Procedural Fluency from Conceptual Understanding ETP

Q: How could you find the roots of $P(x)$ if you didn't recognize the identity $x^3 + 3x^2y + 3xy^2 + y^3 = (x + y)^3$?

[Use a calculator to graph the function $P(x)$ and see where the graph appears to cross the x-axis. Then use synthetic division to test whether that value is a root and find another factor in order to simplify $P(x)$.]

Try It! Answers

5. a. $x = -3$, $x = 2$

b. $x = -1$, $x = 0$, $x = \sqrt{2}i$, $x = -\sqrt{2}i$

Elicit and Use Evidence of Student Thinking ETP

Q: In part a, what do you notice about both sides of the equation?

[Both sides of the equation have an x^3-term, which you can subtract from both sides before solving.]

EXAMPLE 6 Solve a Polynomial Inequality by Graphing

Use and Connect Mathematical Representations ETP

Q: Why is the solution of the inequality the same as the intervals where the graph of $P(x)$ is below the x-axis?

[The inequality is $x^3 - 16x < 0$. You are trying to find the values of $x^3 - 16x$ that are less than zero. The graph of $P(x)$ shows all values of $x^3 - 16x$. The intervals where the graph is below the x-axis are the same as the solutions of $x^3 - 16x < 0$ because the y-value of any point below the x-axis is less than 0.]

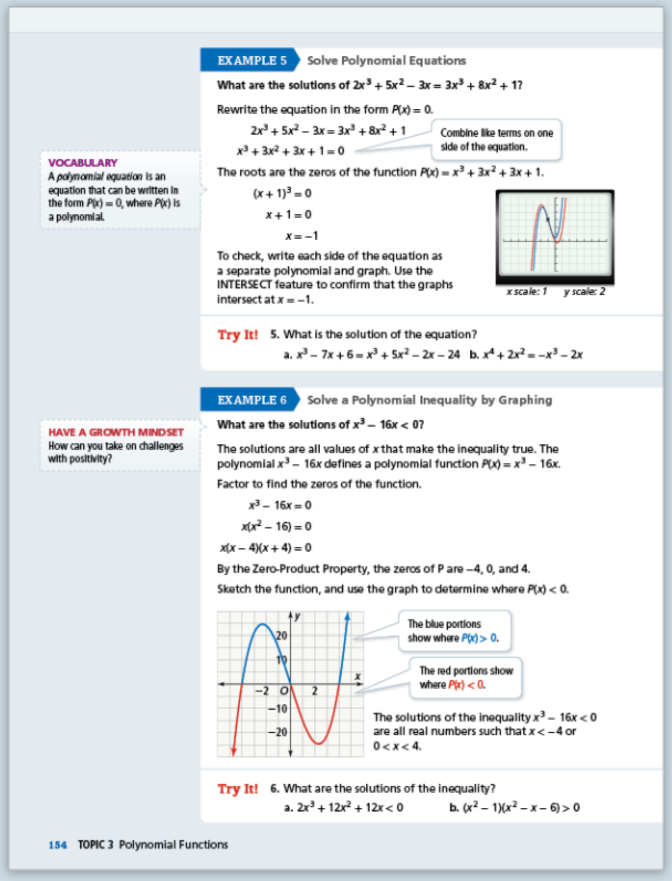
Try It! Answers

6. a. $x < -3 - \sqrt{3}$ or $-3 + \sqrt{3} < x < 0$

b. $x < -2$ or $-1 < x < 1$ or $x > 3$

Have a Growth Mindset: Take on Challenges with Positivity When you struggle with a challenge, you build connections in your brain. Ask: What can you tell yourself so you keep a positive attitude and don't give up when you struggle? What can you do when something seems hard?

Examples are available at SavvasRealize.com.



EXAMPLE 5 Solve Polynomial Equations

What are the solutions of $2x^3 + 5x^2 - 3x = 3x^3 + 8x^2 + 1$?

Rewrite the equation in the form $P(x) = 0$.

$$2x^3 + 5x^2 - 3x = 3x^3 + 8x^2 + 1$$
$$x^3 + 3x^2 + 3x + 1 = 0$$

Combine like terms on one side of the equation.

The roots are the zeros of the function $P(x) = x^3 + 3x^2 + 3x + 1$.

$$(x + 1)^3 = 0$$
$$x + 1 = 0$$
$$x = -1$$

To check, write each side of the equation as a separate polynomial and graph. Use the INTERSECT feature to confirm that the graphs intersect at $x = -1$.

x scale: 1 *y scale: 2*

VOCABULARY
A *polynomial equation* is an equation that can be written in the form $P(x) = 0$, where $P(x)$ is a polynomial.

Try It! 5. What is the solution of the equation?

a. $x^3 - 7x + 6 = x^3 + 5x^2 - 2x - 24$ b. $x^4 + 2x^2 = -x^3 - 2x$

EXAMPLE 6 Solve a Polynomial Inequality by Graphing

What are the solutions of $x^3 - 16x < 0$?

The solutions are all values of x that make the inequality true. The polynomial $x^3 - 16x$ defines a polynomial function $P(x) = x^3 - 16x$.

Factor to find the zeros of the function.

$$x^3 - 16x = 0$$
$$x(x^2 - 16) = 0$$
$$x(x - 4)(x + 4) = 0$$

By the Zero-Product Property, the zeros of P are −4, 0, and 4.

Sketch the function, and use the graph to determine where $P(x) < 0$.

The blue portions show where $P(x) > 0$.

The red portions show where $P(x) < 0$.

The solutions of the inequality $x^3 - 16x < 0$ are all real numbers such that $x < -4$ or $0 < x < 4$.

HAVE A GROWTH MINDSET
How can you take on challenges with positivity?

Try It! 6. What are the solutions of the inequality?

a. $2x^3 + 12x^2 + 12x < 0$ b. $(x^2 - 1)(x^2 - x - 6) > 0$

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Use with EXAMPLES 5 & 6

HABITS OF MIND

Use Structure How does solving $2x^3 + 12x^2 + 12x = 0$ help you to solve the inequality $2x^3 + 12x^2 + 12x < 0$?

[The solutions of the equation help you sketch the related function so you can see where the values of the function are negative.]

RtI Support Struggling Students

USE WITH EXAMPLE 6 Students may struggle with solving cubic inequalities.

Have students practice solving quadratic inequalities.

1. What values of x are solutions to the inequality $x^2 - 12x + 35 < 0$?
[$5 < x < 7$]
2. What values of x are solutions to the inequality $2x^2 - 5x - 3 > 0$?
[$x < -\frac{1}{2}$ or $x > 3$]
3. What values of x are solutions to the inequality $9x^2 - 9x - 4 < 0$?
[$-\frac{1}{3} < x < \frac{4}{3}$]
4. What values of x are solutions to the inequality $x^2 + 9x + 18 > 0$?
[$x < -6$ or $x > -3$]

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Presentation
Practice

CONCEPT SUMMARY Zeros of Polynomials

Q: How does the graph shown verify that a, b, and c are zeros of the function?
[When x is equal to a, b, or c, the graph of the function intersects the x-axis.]

Do You UNDERSTAND? | Do You KNOW HOW?

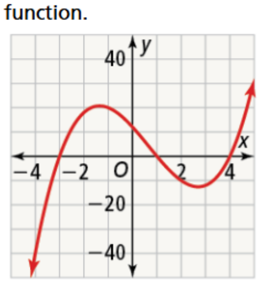
Common Error

Exercise 5 Students may state that the domain is all real numbers. Encourage students to consider whether this result makes sense in the context of this problem. Remind them that even though the function $P(x)$ is greater than zero when $x < -1$, values of x that are less than zero do not make sense in the context of this problem.

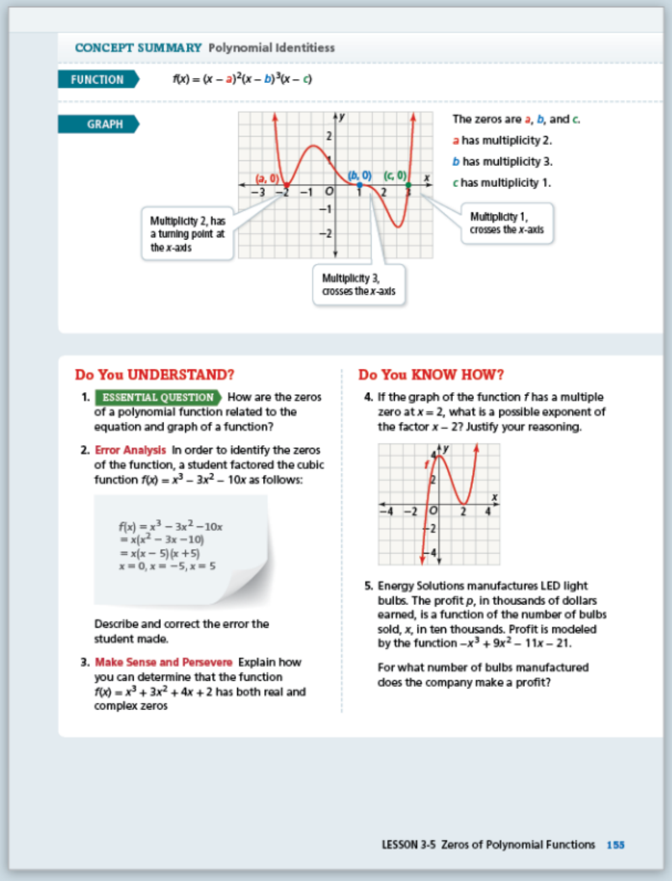
Answers

1. The zeros of polynomial functions show where the polynomial either crosses or touches the x-axis. The sign of the values of the function between the intervals created by the zeros determines whether the function is above or below the x-axis. Additionally, the signs of the values of the function to the left and right of the least and greatest zeros determine the end behavior of the graph.
2. The student factored incorrectly. The zeros are 0, 5, and -2.
3. Graph the function using a graphing calculator or graphing program. Use the graph to estimate one of the zeros of the function. Use synthetic division to confirm that value is a zero and to factor the polynomial. Use the Quadratic Formula to find the remaining zeros.
4. The graph touches, but does not cross, the x-axis at $x = 2$, the exponent is an even number (multiple of 2).
5. $P(x) > 0$ for $3 < x < 7$, so the company should make between 30,000 and 70,000 LED light bulbs.
6. You can use a graphing calculator to check the sketch.
7. Tonya neglected to use synthetic division to factor the polynomial in order to find its complex roots, $-1 + 2i$ and $-1 - 2i$.
8. Graph them simultaneously on the graphing calculator to see if they coincide.
9. When a graph has a multiplicity of a zero that is even, the graph only touches the x-axis, and turns back without crossing. That never occurs in the graph of this polynomial function.

12.



Concept Summary, Do You Understand, and Do You Know How are available at SavvasRealize.com.



CONCEPT SUMMARY Polynomial Identitiess

FUNCTION $f(x) = (x - a)^2(x - b)^3(x - c)$

GRAPH

The zeros are a, b, and c.
a has multiplicity 2.
b has multiplicity 3.
c has multiplicity 1.

Multiplicity 2, has a turning point at the x-axis
Multiplicity 3, crosses the x-axis
Multiplicity 1, crosses the x-axis

Do You UNDERSTAND?

1. ESSENTIAL QUESTION How are the zeros of a polynomial function related to the equation and graph of a function?
2. Error Analysis In order to identify the zeros of the function, a student factored the cubic function $f(x) = x^3 - 3x^2 - 10x$ as follows:
$f(x) = x^3 - 3x^2 - 10x$
$= x(x^2 - 3x - 10)$
$= x(x - 5)(x + 5)$
$x = 0, x = -5, x = 5$
Describe and correct the error the student made.
3. Make Sense and Persevere Explain how you can determine that the function $f(x) = x^3 + 3x^2 + 4x + 2$ has both real and complex zeros

Do You KNOW HOW?

4. If the graph of the function f has a multiple zero at $x = 2$, what is a possible exponent of the factor $x - 2$? Justify your reasoning.
5. Energy Solutions manufactures LED light bulbs. The profit p, in thousands of dollars earned, is a function of the number of bulbs sold, x, in ten thousands. Profit is modeled by the function $-x^3 + 9x^2 - 11x - 21$.
For what number of bulbs manufactured does the company make a profit?

LESSON 3-5 Zeros of Polynomial Functions 155

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Practice

PRACTICE & PROBLEM SOLVING

Practice & Problem Solving and **video tutorials** are available at SavvasRealize.com.

Lesson Practice You may opt to have students complete the automatically scored Practice and Problem Solving items online powered by MathXL for School.

Choose from: **Lesson Practice**

Adaptive Practice

You may also take advantage of the bank of exercises for assigning additional practice.

Assignment Guide

On-level	Advanced
6, 7, 9, 12–30	6–30

Engage Through Student Choice

Promote student agency by allowing students to choose practice items. You may structure this choice in many ways.

For example:

Assign each section a point value. Students choose at least one item from each section and items chosen should have a minimum of 20 total points.

Understand, Apply....................2 points each
Practice....................................1 point each
Assessment Practice................1 point each
Performance Task ....3 points

Item Analysis

Example	Items	DOK
1	12–13, 29	1
	6	2
2	14–16	1
	7, 8, 10	2
3	9, 17, 28	1
4	18, 26	2
	25, 27, 30	3
5	11, 19–21	1
6	22–24	1

Answers

6–9, 12. See answers on previous page.

Scan the QR code to access a video tutorial.

PRACTICE & PROBLEM SOLVING

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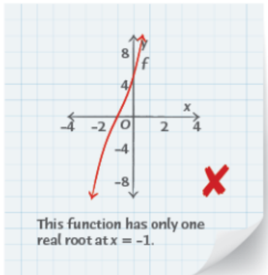
UNDERSTAND

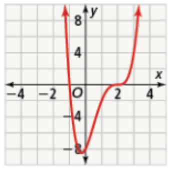
6. Reason If you use zeros to sketch the graph of a polynomial function, how can you verify that your graph is correct?

7. Error Analysis Describe and resolve two errors that Tonya may have made in finding all the roots of the polynomial function, $f(x) = x^3 + 3x^2 + 7x + 5$.



8. Higher Order Thinking How could you use your graphing calculator to determine that $f(x) = (x + 2)(x + 6)(x - 1)$ is not the correct factorization of $f(x) = x^3 + 7x^2 + 16x + 12$? Explain.

9. Generalize How can you determine that the polynomial function shown does not have any zeros with even multiplicity? Explain.



10. Use Structure Factor the polynomial $x^4 - 16$. How many real zeros does the function $g(x) = x^4 - 16$ have? two real zeros at 2 and −2

11. At what points do the graphs of $f(x) = x^3 - 2x^2 - 16x + 20$ and $g(x) = -12$ intersect?
$(-4, -12)$, $(2, -12)$, $(4, -12)$

PRACTICE

Sketch the graph of the function by finding the zeros. SEE EXAMPLE 1

12. $f(x) = 3x^3 - 9x^2 - 12x$

13. $g(x) = (x + 3)(x - 1)(x - 4)$

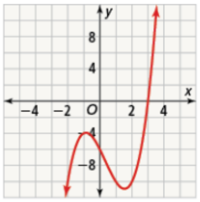
Find the zeros of the function, and describe the behavior of the graph at each zero. SEE EXAMPLE 2

14. $f(x) = x^3 - 8x^2 + 16x$ 0, 4; The graph crosses the x-axis at 0, and it touches the x-axis at 4.

15. $g(x) = x^3 - x^2 - 25x + 25$ −5, 1, 5; The graph crosses the x-axis at −5, 1, and 5.

16. $f(x) = 9x^4 - 40x^2 + 16$ $\pm\frac{2}{3}$, ±4; The graph crosses the x-axis at each zero.

17. What are all the real and complex zeros of the polynomial function shown in the graph? SEE EXAMPLE 3



$f(x) = x^3 - x^2 - 4x - 6$

The zeros of the polynomial function are 3, $-1 + i$, and $-1 - i$.

18. Waterworks is a company that manufactures and sells paddleboards. Their profit P, in hundreds of dollars earned, is a function of the number of paddleboards sold x, measured in thousands. Profit is modeled by the function $P(x) = -3x^3 + 48x^2 - 144x$. What do the zeros of the function tell you about the number of paddleboards that Waterworks should produce? SEE EXAMPLE 4

What are the solution(s) of the equation? SEE EXAMPLE 5

19. $-3x^3 - x^2 + 54x - 40 = 2x^2 + 6x + 20$
$x = -5, x = 2$

20. $2x^3 + 3x^2 - 36 = x^3 - x^2 + 9x$
$x = -4, x = -3, x = 3$

21. $-5x^4 + 4x^2 - 12x = -6x^4 + 3x^3$
$x = 0, x = 3, x = 2i, x = -2i$

What are the solutions of the inequality? SEE EXAMPLE 6

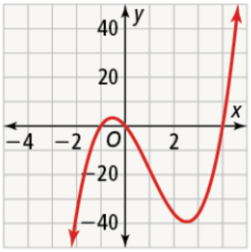
22. $x^3 - 9x > 0$
$-3 < x < 0$ or $x > 3$

23. $0 > 4x^3 + 8x^2 - x - 2$
$x < -2$ or $-\frac{1}{2} < x < \frac{1}{2}$

24. $64x^2 > -4x^3 - x - 16$
$x > -16$

156 TOPIC 3 Polynomial Functions

13.



18. When $P(x)$ is positive, the company earns a profit. $P(x) > 0$ for $4 < x < 12$, so the company should make between 4,000 and 12,000 paddleboards.

Answers

25. **a.** The domain is $0 < t \leq 10$.

b. The zeros are 0 and 10. They represent the times when the projectile is on the ground.

c. The vertex is (5, 122.5), which means at $t = 5$ seconds, the height of the projectile was 122.5 meters.

26. Answers may vary. Sample: The baseball hits the ground a little past 2 seconds after being thrown. However, it is not being thrown from ground level, but from a height of 6.5 feet.

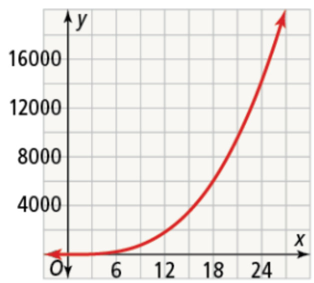
27. **a.** $x(x^2 + 2x - 3) = x(x + 3)(x - 1)$

b. The zeros are $x = 0$, $x = -3$, and $x = 1$.

c. $(x - 1)$ represents the height, and $(x + 3)$ represents the length.

d. width = 2 ft, height = 1 ft, and length = 5 ft.

30. **a.** **Part A**



Part B Venetta did not make a profit during the first two years.

Part C The predicted amount of profit for 2020 is \$828,000.

Practice & Problem Solving is available at SavvasRealize.com.

PRACTICE & PROBLEM SOLVING

APPLY

25. **Make Sense and Persevere** A firework is launched vertically into the air. Its height in meters is given by the function shown, where t is measured in seconds.

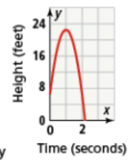
$h = -4.9t^2 + 49t$

a. What is a reasonable domain of the function?

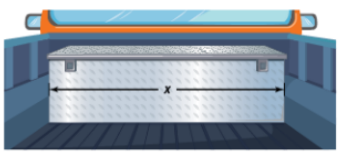
b. What are the zeros of the function? Explain what they represent in this situation.

c. Use technology to find the vertex. What does it represent in this situation?

26. The height of a baseball thrown in the air can be modeled by the function $h(t) = -16t^2 + 32t + 6.5$, where $h(t)$ represents the height in feet of the baseball after t seconds. Explain why the graph of this function only shows one zero.



27. **Model With Mathematics** The height of a rectangular storage box is less than both its length and width. The function $f(x) = x^3 + 2x^2 - 3x$ represents the volume of the rectangular box, where x represents the width of the box, in feet.



a. Find the factored form of $f(x)$.

b. Find the zeros of the function.

c. You know x represents the width of the box. What do the other two factors represent?

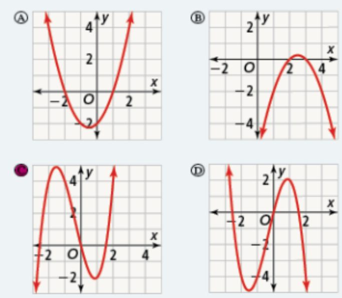
d. Find the dimensions of the box when the volume is 10 ft^3.

ASSESSMENT PRACTICE

28. Complete each statement so it means the same as 4 is *a zero of the function*.

The graph of the function crosses the <u>x-axis</u> at 4. <u>$(x - 4)$</u> is a factor of the polynomial.

29. **SAT/ACT** Without the use of a graphing calculator, determine which of the following functions is the graph of $f(x) = x^3 + x^2 - 4x$.



30. **Performance Task** Venetta opened several deli sandwich franchises in 2000. The profit P (in hundreds of dollars) of the franchises in t years (since the franchises opened) can be modeled by the function $P(t) = t^3 + t^2 - 6t$.

Part A Sketch a graph of the function.

Part B Based on the model, during what years did Venetta not make a profit?

Part C If the model is appropriate, predict the amount of profit Venetta will receive from her franchises in 2020.

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Assessment

LESSON QUIZ

Use the Lesson Quiz to assess students' understanding of the mathematics in the lesson.

Students can take the Lesson Quiz online or you can download a printable copy from **SavvasRealize.com.** The Lesson Quiz is also available in the *Assessment Sourcebook.*

Item Analysis

Item	DOK	Skills Review and Practice
1A	2	F62
1B	2	A73
2	2	F62
3	2	F62
4	1	F62
5	2	F62

RtI Use the student scores on the Lesson Quiz to prescribe differentiated assignments.

If students take the Lesson Quiz online, it will be automatically scored and appropriate differentiated practice will be assigned based on student performance.

I Intervention	0–3 points	• Reteach to Build Understanding • Mathematical Literacy and Vocabulary • Lesson Virtual Nerd videos
O On-Level	4 points	• Enrichment
A Advanced	5 points	• Enrichment

Lesson Quiz is available at SavvasRealize.com.

ASSESSMENT RESOURCES

Name

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3-5 Lesson Quiz

Zeros of Polynomial Functions

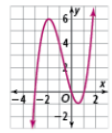
1. **Part A**

What are all the zeros of $f(x) = x^3 + 2x^2 - 3x$?

Zeros: -3, 0, 1

Part B

Use the zeroes to sketch the graph of f.



2. What x-values are solutions of $x^3 + 5x^2 - x - 7 = x^2 + 6x + 3$? Simplify the polynomial and find the zeros. Complete the sentence.

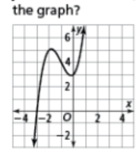
The zeros are at $x = -5$, $x = -1$, and $x = 2$.

3. What values of x are solutions of the inequality $x^3 - 4x > 0$?

The inequality is true for

☐ $0 < x < 2$ ■ $-2 < x < 0$ ☐ $x < -2$ ☐ $x < 2$	and	☐ $0 < x < 2$. ☐ $x > 0$. ■ $x > 2$. ☐ $x > 4$.

4. What are the zeros and their multiplicities for the function $y = x^3 + 3x^2 + x + 3$, shown in the graph?



Ⓐ -3, -1, and 1, each multiplicity 1
Ⓑ -3, multiplicity 1
Ⓒ 3, multiplicity 2
Ⓓ -3, multiplicity 3

5. Find the zeros of $f(x) = -x^3 - 2x^2 + 7x - 4$. Then describe the behavior of the graph of f at each zero.

Ⓐ 4, -1; As $x \to -\infty$, $f \to -\infty$. When $-1 < x < 4$, $f < 0$. At $x = 4$, f is tangent to the x-axis, so when $x > 1$, $f \to \infty$.
Ⓑ -4, 1; As $x \to -\infty$, $f \to -\infty$. When $-4 < x < 1$, $f > 0$. At $x = 1$, f is tangent to the x-axis, so when $x > 1$, $f \to -\infty$.
Ⓒ 4, -1; As $x \to -\infty$, $f \to \infty$. When $-1 < x < 4$, $f > 0$. At $x = 4$, f is tangent to the x-axis, so when $x > 1$, $\to \infty$.
Ⓓ -4, 1; As $x \to -\infty$, $f \to \infty$. When $-4 < x < 1$, $f < 0$. At $x = 1$, f is tangent to the x-axis, so when $x > 1$, $f \to -\infty$.

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Practice

DIFFERENTIATED RESOURCES

I = Intervention O = On-Level A = Advanced

Reteach to Build Understanding I

Provides scaffolded reteaching for the key lesson concepts.

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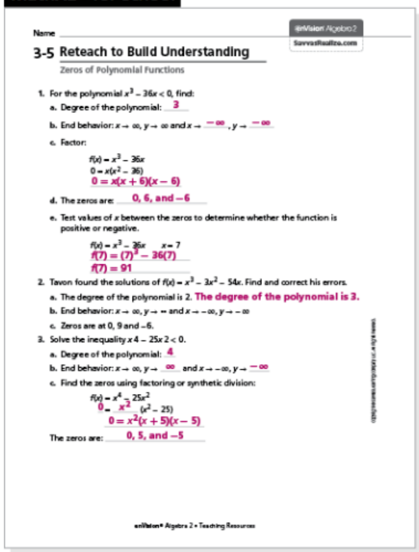
Name

3-5 Reteach to Build Understanding

Zeros of Polynomial Functions

1. For the polynomial $x^3 - 36x < 0$, find:

a. Degree of the polynomial: 3

b. End behavior: $x \to \infty, y \to \infty$ and $x \to -\infty, y \to -\infty$

c. Factor:

$f(x) = x^3 - 36x$

$0 = x(x^2 - 36)$

$0 = x(x + 6)(x - 6)$

d. The zeros are: 0, 6, and −6

e. Test values of x between the zeros to determine whether the function is positive or negative.

$f(x) = x^3 - 36x$ $x = 7$

$f(7) = (7)^3 - 36(7)$

$f(7) = 91$

2. Tavon found the solutions of $f(x) = x^3 - 3x^2 - 54x$. Find and correct his errors.

a. The degree of the polynomial is 2. The degree of the polynomial is 3.

b. End behavior: $x \to \infty, y \to \infty$ and $x \to -\infty, y \to -\infty$

c. Zeros are at 0, 9 and −6.

3. Solve the inequality $x^4 - 25x^2 < 0$.

a. Degree of the polynomial: 4

b. End behavior: $x \to \infty, y \to \infty$ and $x \to -\infty, y \to -\infty$

c. Find the zeros using factoring or synthetic division:

$f(x) = x^4 - 25x^2$

$0 = x^2(x^2 - 25)$

$0 = x^2(x + 5)(x - 5)$

The zeros are: 0, 5, and −5

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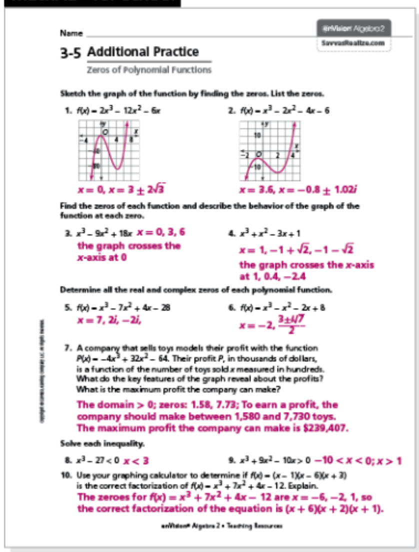
Additional Practice I O

Provides extra practice for each lesson.

MathXL® for School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3-5 Additional Practice

Zeros of Polynomial Functions

Sketch the graph of the function by finding the zeros. List the zeros.

1. $f(x) = 2x^3 - 12x^2 - 6x$

$x = 0, x = 3 \pm 2\sqrt{3}$

2. $f(x) = x^3 - 2x^2 - 4x - 6$

$x = 3.6, x = -0.8 \pm 1.02i$

Find the zeros of each function and describe the behavior of the graph of the function at each zero.

3. $x^3 - 9x^2 + 18x$ $x = 0, 3, 6$ the graph crosses the x-axis at 0

4. $x^3 + x^2 - 3x + 1$ $x = 1, -1 + \sqrt{2}, -1 - \sqrt{2}$ the graph crosses the x-axis at 1, 0.4, −2.4

Determine all the real and complex zeros of each polynomial function.

5. $f(x) = x^3 - 7x^2 + 4x - 28$ $x = 7, 2i, -2i$

6. $f(x) = x^3 - x^2 - 2x + 8$ $x = -2, \frac{3 \pm i\sqrt{7}}{2}$

7. A company that sells toys models their profit with the function $P(x) = -4x^3 + 32x^2 - 64$. Their profit P, in thousands of dollars, is a function of the number of toys sold x measured in hundreds. What do the key features of the graph reveal about the profits? What is the maximum profit the company can make?

The domain > 0; zeros: 1.58, 7.73; To earn a profit, the company should make between 1,580 and 7,730 toys. The maximum profit the company can make is $239,407.

Solve each inequality.

8. $x^3 - 27 < 0$ $x < 3$

9. $x^3 + 9x^2 - 10x > 0$ $-10 < x < 0; x > 1$

10. Use your graphing calculator to determine if $f(x) = (x - 1)(x - 6)(x + 2)$ is the correct factorization of $f(x) = x^3 + 7x^2 + 4x - 12$. Explain.

The zeros for $f(x) = x^3 + 7x^2 + 4x - 12$ are $x = -6, -2, 1$, so the correct factorization of the equation is $(x + 6)(x + 2)(x + 1)$.

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Enrichment O A

Presents engaging problems and activities that extend the lesson concepts.

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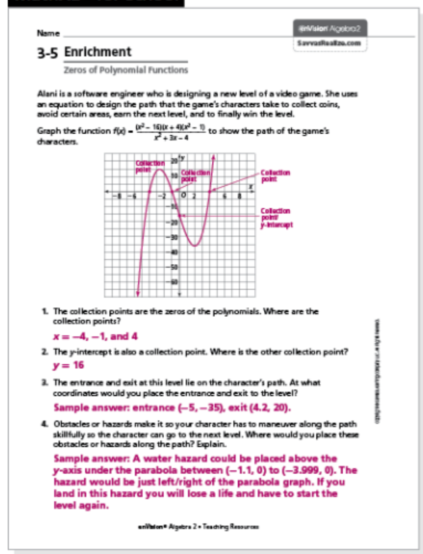
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3-5 Enrichment

Zeros of Polynomial Functions

Alani is a software engineer who is designing a new level of a video game. She uses an equation to design the path that the game's characters take to collect coins, avoid certain areas, earn the next level, and to finally win the level.

Graph the function $f(x) = \frac{(x^2 - 16)(x + 4)(x^2 - 1)}{x^2 + 3x - 4}$ to show the path of the game's characters.

1. The collection points are the zeros of the polynomials. Where are the collection points?

$x = -4, -1$, and 4

2. The y-intercept is also a collection point. Where is the other collection point?

$y = 16$

3. The entrance and exit at this level lie on the character's path. At what coordinates would you place the entrance and exit to the level?

Sample answer: entrance (−5, −35), exit (4.2, 20).

4. Obstacles or hazards make it so your character has to maneuver along the path skillfully so the character can go to the next level. Where would you place these obstacles or hazards along the path? Explain.

Sample answer: A water hazard could be placed above the y-axis under the parabola between (−1.1, 0) to (−3.999, 0). The hazard would be just left/right of the parabola graph. If you land in this hazard you will lose a life and have to start the level again.

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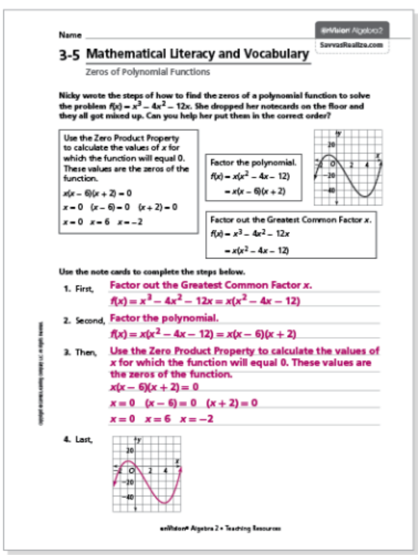
Mathematical Literacy and Vocabulary I O

Helps students develop and reinforce understanding of key terms and concepts.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3-5 Mathematical Literacy and Vocabulary

Zeros of Polynomial Functions

Nicky wrote the steps of how to find the zeros of a polynomial function to solve the problem $f(x) = x^3 - 4x^2 - 12x$. She dropped her notecards on the floor and they all got mixed up. Can you help her put them in the correct order?

Use the Zero Product Property to calculate the values of x for which the function will equal 0. These values are the zeros of the function. $x(x - 6)(x + 2) = 0$; $x = 0$ $(x - 6) = 0$ $(x + 2) = 0$; $x = 0$ $x = 6$ $x = -2$

Factor the polynomial. $f(x) = x(x^2 - 4x - 12) = x(x - 6)(x + 2)$

Factor out the Greatest Common Factor x. $f(x) = x^3 - 4x^2 - 12x = x(x^2 - 4x - 12)$

Use the note cards to complete the steps below.

1. First, Factor out the Greatest Common Factor x. $f(x) = x^3 - 4x^2 - 12x = x(x^2 - 4x - 12)$

2. Second, Factor the polynomial. $f(x) = x(x^2 - 4x - 12) = x(x - 6)(x + 2)$

3. Then, Use the Zero Product Property to calculate the values of x for which the function will equal 0. These values are the zeros of the function. $x(x - 6)(x + 2) = 0$; $x = 0$ $(x - 6) = 0$ $(x + 2) = 0$; $x = 0$ $x = 6$ $x = -2$

4. Last,

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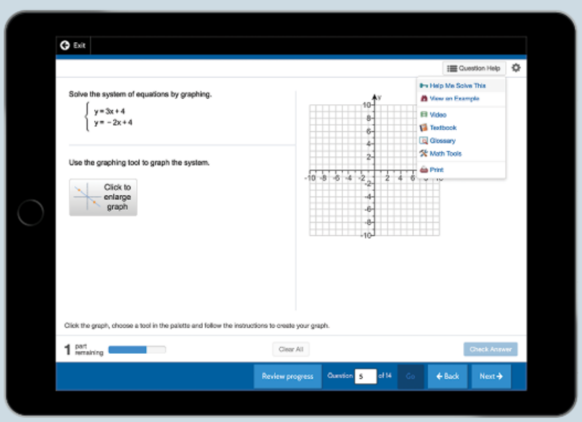
Digital Differentiated Resources

The **Reteach to Build Understanding**, **Additional Practice**, and **Enrichment** activities are available as digital assignments. These activities are automatically assigned when students complete the lesson quiz online and are automatically scored.



TOPIC 3

Mathematical Modeling in 3 Acts: What Are the Rules?

SavvasRealize.com

Video / Audio

Lesson Overview

FOCUS

Objective

Students will be able to:

- ✔ Use mathematical modeling to represent a problem situation and to propose a solution.
- ✔ Test and verify the appropriateness of their math models.
- ✔ Explain why the results from their mathematical models might not align exactly with the problem situation.

Essential Understanding

Many real-world problem situations can be represented with a mathematical model, but that model might not represent the real-world situation exactly.

COHERENCE

Earlier in this topic, students:

- Operated with polynomials and graphed polynomial functions.

In this lesson, students:

- Develop a mathematical model to represent and propose a solution to a problem situation involving polynomial functions and zeros.

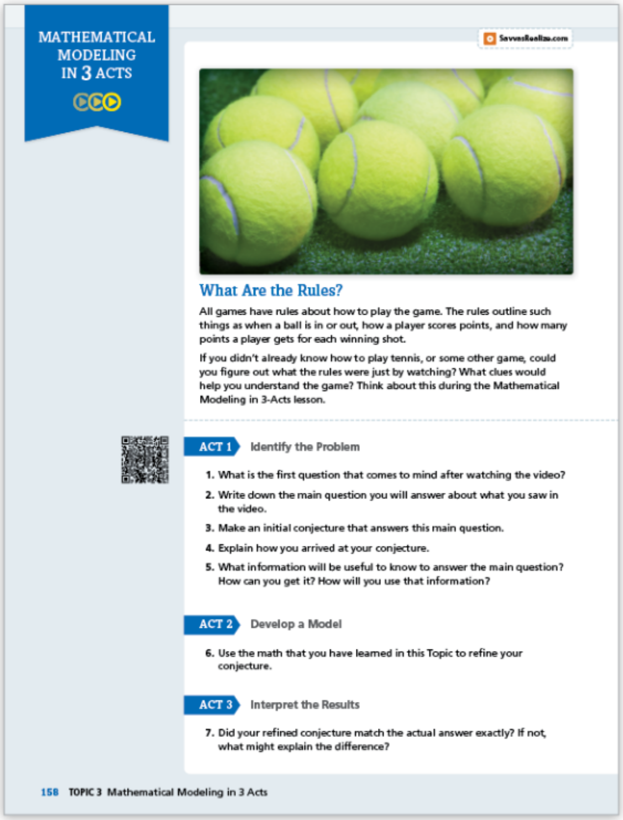
Later in this topic, students will:

- Refine their mathematical modeling skills using polynomial functions.

RIGOR

This mathematical modeling lesson focuses on application of both math content and math practices and processes.

- Students draw on their understanding of concepts related to polynomial functions to develop a representative model.
- Students apply their mathematical model to test and validate its applicability to similar problem situations.



MATHEMATICAL MODELING IN 3 ACTS

SavvasRealize.com

What Are the Rules?

All games have rules about how to play the game. The rules outline such things as when a ball is in or out, how a player scores points, and how many points a player gets for each winning shot.

If you didn't already know how to play tennis, or some other game, could you figure out what the rules were just by watching? What clues would help you understand the game? Think about this during the Mathematical Modeling in 3-Acts lesson.

ACT 1 Identify the Problem

1. What is the first question that comes to mind after watching the video?
2. Write down the main question you will answer about what you saw in the video.
3. Make an initial conjecture that answers this main question.
4. Explain how you arrived at your conjecture.
5. What information will be useful to know to answer the main question? How can you get it? How will you use that information?

ACT 2 Develop a Model

6. Use the math that you have learned in this Topic to refine your conjecture.

ACT 3 Interpret the Results

7. Did your refined conjecture match the actual answer exactly? If not, what might explain the difference?

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Student Companion

Students can do their work for the task in their *Student Companion* or on **SavvasRealize.com.**

Mathematics Overview

The Importance of Modeling

Rather than presenting a problem with all the necessary information, diagrams, and graphs in advance, this task also asks students to participate in the formulation of a problem.

Students decide what questions are interesting, what quantities are important, and how they could access information. They choose which math concepts are relevant to the task, using approximations or assumptions in the solution when necessary.

Finally, the task includes an actual, real-world answer. Students discuss possible sources of error inherent in using math to model real-world situations. Then they evaluate the usefulness of their models in the situation, improving them if necessary.

Mathematical Practices

Model With Mathematic

To solve the problem presented, students identify variables and the relationship among them, develop a model that represents the situation, and use the model to propose a solution. Students interpret their solutions and propose explanations for why their answer may not match the real-world answer.

Other Practices

Students also engage in sense-making, perseverance, and abstract and quantitative reasoning as they complete the task. In testing their models, students look for patterns in their models and use that structure to find a solution.

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Video / Audio

What Are the Rules?

In this mathematical modeling task, students will explore and apply concepts related to polynomial functions and equations. Students will analyze a situation in which a polynomial function affects numbers differently, and they will be tasked with determining why. To do so, they apply concepts that they study in Topic 3.

ACT 1 The Hook

Play the video. The video shows a tennis racquet with a polynomial function and seven balls numbered -3 to 3. A player uses the racquet to hit three balls, one at a time. The first and third balls, 2 and -3, have a negative result, while the second ball, -1, has a positive result.

After the question brainstorming, present to students the Main Question they will be tasked with answering. Remind students to write down their questions and conjectures.

MAIN QUESTION

What will happen for each of the other numbered balls? Why?

ACT 2 Modeling With Math

Think about the task. Ask students to speculate how they could determine why two had a negative result and one had a positive result. How can they figure out what will happen when the player hits each of the remaining balls? Then have them think about what information they would need to answer these questions.

Reveal the information. The images show a polynomial written on the racquet, the results of the three serves from Act 1, and define what makes a legal serve. If students need help seeing the pattern, suggest that they investigate the related polynomial equation $0 = x^3 - 3x^2 - x + 3$.

What's the connection? Give students time to struggle as they think about how their ideas are connected to what they learned in this topic about polynomial functions. Ask them to think about attributes of polynomial functions. As needed, lead them to discuss how to find the linear factors of a function and find its zeros. How will they know that they have found all of them?

INTERESTING MOMENTS WITH STUDENTS

Guess and check creates a need for stronger factoring methods. If one of the zeros had been 20, it could have been tedious to check that many values in the function. If one of the zeros had been 2.5, students might have guessed only integers and missed it entirely.

Necessary Information

Polynomial function

$f(x) = x^3 - 3x^2 - x + 3$

Ball behavior

Illegal serve: 2, -3

Legal serve: -1

The player does not have control over whether each serve is legal or not. Only the number on the ball affects each result.

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Video / Audio

ACT 3 The Solution

Play the video. The final video shows what happens when each of the remaining balls hits the polynomial: the remaining zeros of the polynomial function are revealed. Offer praise to the students whose conjectures match the actual answer.

MAIN QUESTION ANSWER

-2 and 0 will be illegal serves, and 1 and 3 will be legal serves. The factored form of the polynomial $f(x) = x^3 - 3x^2 - x + 3$ is $f(x) = (x + 1)(x - 1)(x - 3)$, so it has zeros at $x = -1$, 1, and 3.

Do the "post-game" analysis. Ask students how they could determine whether they found all of the factors. Students could use the Fundamental Theorem of Algebra. Because it is a polynomial of degree three, we know it has exactly 3 zeros. They could also use factoring, polynomial long division, or synthetic division to write the factored form of the polynomial.

ONE POSSIBLE SOLUTION

Factor by grouping.

$f(x) = x^3 - 3x^2 - x + 3$

$f(x) = (x^3 - 3x^2) + (-x + 3)$

$f(x) = x^2(x - 3) - 1(x - 3)$

$f(x) = (x^2 - 1)(x - 3)$

$f(x) = (x + 1)(x - 1)(x - 3)$

The factored form of the polynomial function is $f(x) = (x + 1)(x - 1)(x - 3)$.

Therefore, by the Zero-Product Property, the zeros are -1, 1, and 3. Balls 1 and 3 will be legal serves, and the others will be illegal.

SEQUEL

As students finish, tell them that the player has another racquet with the polynomial function $g(x) = x^3 - 2x - 5x^2 + 10$ on it. Ask them which balls from the video the player should use now. [Answer: The zeros are $x = \sqrt{2}$, $x = -\sqrt{2}$, and $x = 5$. The player should not use any of the tennis balls in the video.]